

AFRI Terminal - Feature Expansion Specification

What Claude should add to the existing terminal (<http://127.0.0.1:8081/terminal.html>, files in the GitHub repo / D:\mutxri-terminal). Design DNA: Godel Terminal's black+mint aesthetic and command palette, Bloomberg's 3-panel pro layout and function codes, TradingView's charting and heatmap quality. Product name is TBD - use 'AFRI' as placeholder. Every feature below is specified so Claude can build it without further questions. Last updated August 2026.

1. Design Language (Copy These Exactly)

1.1 Color Palette (from Godel's CSS)

Variable	Value	Use
--bg	#000000	Page background
--bg2	#0a0a0a	Panel background
--bg3	#141414	Sub-panel / headers
--bg4	#1f1f1f	Command bar / active
--line	#1a1a1a	Borders
--line2	#262626	Stronger borders
--tx	#f0f0f0	Primary text
--tx2	#9a9a9a	Secondary text
--tx3	#6a6a6a	Muted text
--up	#33e29a	Gains (mint green)
--down	#f87171	Losses (red)
--acc	#33e29a	Accent / active
--acc2	#1ecfb0	Secondary accent (teal)

1.2 Fonts

Google Fonts: Oxygen Mono (ALL data/numbers/tickers/tables), Inter (UI text/labels), Sofia Sans Semi Condensed (logo/brand). Never use JetBrains Mono. Data text is always monospace - this is the Bloomberg feel.

1.3 Layout DNA

Bloomberg-style 3-panel: left = watchlist/command-driven lists, center = chart + heatmap, right = tabbed panels (News/FX/RATES/FUND/Detail). Top bar = brand, exchange tabs, search, market clock. Bottom = status line. Everything dense, keyboard-friendly, monospace-heavy. Scanline-free. Corner radius 0-2px (terminal, not web app). Godel accent: mint highlight on hover/active, thin borders, no shadows except command bar.

2. Feature Priority Overview

Priority	Feature	Why
P0	Command palette overhaul (Bloomberg codes)	Core UX - what makes it a terminal
P0	Screener (EQS/SCREEN)	Explicitly requested, data ready
P0	Morning brief improvements	Already works - make it richer
P1	Multi-chart workspace / watchlist persistence	Pro users expect it
P1	Company announcements feed (REG)	Differentiator - regulators
P1	Candlestick polish + indicators	TradingView-grade charts
P1	Portfolio watchlist with localStorage	Sticky sessions
P2	Bonds/T-bills board (BND)	Income investors
P2	Global commodities (GLCO)	African angle: gold, platinum, cocoa
P2	Analyst ratings aggregation	Where free data allows
P2	Auth + multi-user	Before public deployment
P3	Mobile PWA	African retail is mobile-first
P3	i18n (FR, AR, SW, ZULU)	Reach
P3	Export to Excel/CSV	FA command parity

3. P0 - Command Palette Overhaul (Bloomberg Function Codes)

The backtick palette exists but is thin. Make it a full Bloomberg-style function-code system. Every code must work with TICKER PREFIX (e.g. 'SOL DES', 'COMI GIP') and bare (e.g. 'WEI'). Add a discoverability help screen (HELP) listing every code with a one-line description, rendered as a searchable list.

	Function	Behavior
	Security Description	Full company overview: business, sector, quote, mcap, key stats, 52W range, description from AF/Wiki
	Global Identifier	All identifiers: ticker, ISIN, exchange, currency, listing date, sector, industry
	Chart	Area/line/candles, all ranges, full-screen option
	Quote Panel	Live quote block: bid/ask where available, open/high/low/prev, volume, 52W
VS	Company News	Ticker-filtered news (Google News query 'TICKER company' + AF news)
	Filings	All documents/filings for the security, sorted desc, clickable PDFs
	Financial Analysis	Income statement, balance sheet, cash flow - from AF financial_summary + statement tables where av
DIV / ERN	Fundamentals	Dividends, earnings, per-share metrics, major shareholders
	Options Monitor	JSE options chain (where free data exists); show 'not available' honestly elsewhere
	Time & Sales	Recent trades list - from Yahoo intraday where available
	Holders	Major shareholders (ownership.json) + institutional holders where known
	World Indices	Index board: all 4 exchanges + global (SPX, FTSE, DAX, Nikkei, Hang Seng) via Yahoo
	Earnings Matrix	EPS, latest earnings date, dividend yield across watchlist
SCREEN	Equity Screener	Filter 768 stocks: exchange, sector, price range, % change, volume, dividend yield - sortable table, exp
	Forex	Full FX board (existing + more pairs)
	Global Commodities	Gold, platinum, silver, oil, cocoa, coffee, tea - via Yahoo commodity symbols
	Bonds	T-bill/bond yields per country (curated from central banks)
	Regulatory	Regulator wire: CMA/SEC/FSCA/FRA/CB announcements
	Morning Brief	Generate + email today's brief on demand
	Help	Searchable command reference
	Quote Monitor	Watchlist view

4. P0 - Equity Screener (EQS/SCREEN)

TradingView-grade screener over all 768 stocks. Data is already available: heatmap payload has price/chgPct/volume/sector for every stock; fundamentals.json + ownership.json add dividends and holders.

UI: full-panel screener (center + right combined or overlay). Filter rows: Exchange (multi-select JSE/EGX/NGX/NSE), Sector (dropdown from classified sectors), Price min/max, % change min/max, Volume min, Dividend yield min (from fundamentals dividends / price), Currency. Result table: Ticker, Company, Exchange, Sector, Price, Chg%, Volume, Dividend yield - sortable by any column, click row = open security. Export button (CSV download). Persist last filter set in localStorage. Honest labels: JSE/EGX live prices, NGX/NSE EOD.

5. P0 - Morning Brief Improvements

The existing morning_brief.py works (indices, FX, rates, headlines). Add: 1) TOP MOVERS section - biggest gainers/losers per exchange from live quotes (top 5 each, with % change). 2) DIVIDENDS section - today's/upcoming dividend announcements from AF news + fundamentals. 3) ANNOUNCEMENTS section - latest corporate announcements. 4) MARKET CALENDAR - today's expected events (earnings dates where known). 5) Custom HTML template with the mint theme, fully responsive for phone reading. 6) Optional second recipient (investor partner). 7) Weekend handling: skip Saturday (no fresh data), send a lighter 'week ahead' brief on Monday.

6. P1 - Charts (TradingView-Grade)

Current charts use lightweight-charts with AREA/LINE/CANDLE. Upgrade to TradingView quality:

Feature	Spec
Indicators	MA(20/50/200), EMA, RSI(14), MACD, Bollinger Bands - toggles in chart tools, drawn as overlay series
Timeframes	1D intraday (where Yahoo provides), 1W, 1M, 3M, 6M, 1Y, 2Y, 5Y, MAX
Fullscreen	F key or button expands chart to full viewport
Crosshair data readout	OHLC + volume + % change at crosshair position (top-left readout like TradingView)
Multi-pane	Price pane + volume pane + indicator pane (MACD/RSI below)
Chart settings	Colors, grid on/off, log scale toggle (saved to localStorage)
Compare	Overlay a second symbol (e.g. SOL vs J203 index) - "COMPARE" command
Export	Download chart as PNG

Data note: Yahoo chart API returns daily bars reliably; intraday 1m/5m bars work for JSE/EGX during session but are rate-limited - use daily for 1M+ ranges, intraday only on demand.

7. P1 - Watchlist & Workspace Persistence

1) localStorage watchlist: user adds symbols (star icon or ADD command), stored per-browser, rendered as a pinned section above the exchange listing. 2) Layout persistence: active exchange tab, active right tab, chart type, ranges - all restored on reload. 3) Multi-chart workspace: allow opening a second chart panel (split view) for compare. 4) Keyboard: arrow keys navigate watchlist, Enter opens, Esc closes command bar, F fullscreen, W watchlist toggle.

8. P1 - REG - Regulatory Wire

A scrolling feed of official regulatory/central-bank releases, aggregated via Google News RSS queries (verified reachable): CMA Kenya (cma.or.ke), SEC Nigeria (sec.gov.ng), FSCA South Africa (fsca.co.za), FRA Egypt (fra.gov.eg), CBK, CBN, SARB, CBE, National Treasuries. Queries: 'site:or.gov.ke OR site:sec.gov.ng OR site:fsca.co.za' etc + institution name queries. Each item: source, headline, date, link. New REG tab in right panel + REG command. This is the trust differentiator - no African retail terminal aggregates regulators.

9. P2 - Additional Panels

9.1 GLCO - Global Commodities

Board of commodities relevant to Africa via Yahoo symbols: Gold (GC=F), Silver (SI=F), Platinum (PL=F), Brent (BZ=F), WTI (CL=F), Cocoa (CC=F), Coffee (KC=F), Sugar (SB=F), Corn (ZC=F), Wheat (ZW=F). Table: commodity, price, day %, chart sparkline (mini area). Note: African commodity angle = gold (SA), platinum (SA), cocoa (GH/CI), coffee (KE/ET/UG), tea (KE).

9.2 BND - Bonds & T-Bills

Curated bond/t-bill yield table per country: KE T-bills (91/182/364 day), NG FGN bonds, ZA R186/R203 government bonds, EG T-bills. Sources: central bank auction results (curated, updated on announcements). Show: instrument, maturity, yield, last updated, source. Honest note where delayed.

9.3 Analyst Ratings

Where free data allows: aggregate analyst targets/ratings from public broker notes surfaced via Google News ('TICKER company broker rating target price when:30d'). Display: consensus (Buy/Hold/Sell), latest target, source links. Honest note: full consensus needs licensed data (Refinitiv/Bloomberg).

9.4 Time & Sales (TAS)

Recent trades for JSE/EGX using Yahoo intraday chart data (last ~100 bars, 1m interval, session only). Table: time, price, volume, cumulative. Honest label when market closed (show last session).

10. P2 - Auth & Deployment

Before public launch: 1) Simple auth (email+password or magic link) with sessions - free options: Cloudflare Access, or lightweight JWT in the Python server with a users.json store. 2) Rate limiting per user on Yahoo proxy (fair-use cache already exists - add per-IP limits). 3) Deployment: free tier options - Railway/Render/Cloudflare Workers for the server (Python server = Railway/Render), frontend static. 4) HTTPS via the host. 5) Honest data-license labels on every panel (LIVE / 15-min delay / EOD / curated).

11. P3 - Mobile & International

1) PWA: manifest + service worker, installable, offline shell showing last-loaded data. African retail invests on phones - this is a reach play. 2) Mobile layout: single column - top = quote + mini chart, then tabs. 3) Localization: FR (francophone Africa), AR (Egypt), SW (East Africa), ZU (SA). Start with FR + AR. 4) Currency toggle: show prices in local currency AND USD equivalent. 5) Alerts: price-above/below threshold alerts (in-app first, email later).

12. Data Sources (verified working - use these)

Need	Source	Notes
Quotes JSE/EGX	Yahoo chart API	.JO and .CA suffix, live, cached 240s
Dividends JSE/EGX	Yahoo chart events=div	Full history, cached 24h
Indices	Yahoo (^J203.JO, ^EGX30.CA, ^GSPC...)	Live
FX	Yahoo (USDZAR=X etc)	Live, 6+ pairs
Commodities	Yahoo (GC=F, PL=F, CC=F...)	Live
US yields	Yahoo (^TNX, ^FVX, ^IRX, ^TYX)	Live
NGX/NSE EOD	African Financials share-price tables	Daily EOD, scraped
Fundamentals/filings/dividends NGX/NSE	African Financials company pages (fundamentals.json)	190 companies
Financial summaries	AF filing meta descriptions (fill_financial_summary.py)	Revenue/EBIT/PAT/EPS/assets
Tickers/sectors	TradingView screener (tradingview-screener pip)	EGX 290, NGX 137, KE 57
News (S&P Global etc)	Google News RSS (7 queries) + 5 direct RSS	25+ sources
Logos	Google favicon (company_domains.json)	172 domains
Central bank rates	Curated (CB_RATES in afri_server.py)	Updated on rate decisions
Ownership	ownership.json (curated)	26 liquid names, extensible

DEAD ENDS (verified - do not retry): Mubasher API (403), EGX site (TSPD challenge), investing.com (403), Yahoo quoteSummary/fundamentals (401), Wikipedia EGX tables (none), Zawya (404), StockAnalysis (no JSE/EGX).

13. Architecture Notes For Claude

Current: single-file terminal.html (all CSS/JS inline, lightweight-charts + Google Fonts CDN) + afri_server.py (Python stdlib http.server, cache dict, warm thread, /api/* endpoints). Keep this simplicity unless a feature demands more.

Recommended additions: 1) Keep frontend vanilla JS (no build step) - it's the speed advantage. 2) Server: add /api/screener (server-side filtering on stocks.json + cached quotes), /api/commodities, /api/reg. 3) localStorage keys: 'afri_watchlist', 'afri_layout'. 4) Chart indicators: lightweight-charts supports overlay series (addLineSeries for MAs) - no new library needed; RSI/MACD computed client-side from bars. 5) Never block the UI on Yahoo fetches - always progressive fill + cache. 6) Every new data panel must show its source + freshness + license status (LIVE/DELAYED/EOD/CURATED) - this is the trust moat.

Quality bar: no console errors, all panels work with no data (graceful 'no data' states), keyboard navigation consistent, theme variables used everywhere (no hardcoded colors), responsive down to 360px width.

14. Definition Of 'Flawless' (Acceptance Criteria)

Claude's work is done when: 1) Every P0 feature exists and works from a clean browser (no console errors). 2) All charts render area/line/candle with indicators. 3) Screener filters correctly across all 4 exchanges and exports CSV. 4) Morning brief sends and includes top movers + dividends. 5) Watchlist persists across reload. 6) REG feed shows regulator items. 7) Every panel labels its data freshness. 8) The whole thing runs with just python afri_server.py - no pip installs, no build step. 9) Zero hardcoded secrets in the repo. 10) Mobile layout usable. When those pass, the app is 'flawless' by the owner's standard.

END OF SPEC. Claude: build in priority order (P0 first), commit each feature, and never fake data - show honest 'not available on free feed' states instead.